

Experiment-6

Write a JavaScript program to validate form inputs (e.g., Email, empty fields).

```
<!DOCTYPE html>
<html>
<head>
<title>Form Validation</title>
<script>
function validateForm() {

// Get values
var name = document.getElementById("name").value;
var email = document.getElementById("email").value;

// Check empty fields
if (name == "") {
alert("Name field cannot be empty");
return false;
}

if (email == "") {
alert("Email field cannot be empty");
return false;
}

// Email validation pattern
var emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

if (!email.match(emailPattern)) {
alert("Please enter a valid email address");
return false;
}

alert("Form submitted successfully!");
return true;
}
</script>
</head>

<body>

<h2>Registration Form</h2>

<form onsubmit="return validateForm()">

Name: <input type="text" id="name"><br><br>
```

Email: <input type="text" id="email">

<input type="submit" value="Submit">

</form>

</body>

</html>

Experiment-7

Create a web page that uses JavaScript to display dynamic content using DOM.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Dynamic DOM Example</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial;
```

```
text-align: center;
```

```
margin-top: 50px;
```

```
}
```

```
button {
```

```
padding: 8px 15px;
```

```
font-size: 16px;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2 id="heading">Welcome!</h2>
```

```
<button onclick="changeContent()">Click Me</button>
```

```
<button onclick="addParagraph()">Add Message</button>
```

```
<div id="content"></div>
```

```
<script>
```

```

function changeContent() {
// Change heading text
document.getElementById("heading").innerHTML = "Hello, JavaScript DOM!";

// Change heading color
document.getElementById("heading").style.color = "blue";
}

function addParagraph() {
// Create new paragraph element
var para = document.createElement("p");

// Add text to paragraph
para.innerHTML = "This is dynamically added content.";

// Append paragraph to div
document.getElementById("content").appendChild(para);
}
</script>

</body>
</html>

```

Experiment-8

Write a JavaScript program for a simple calculator using functions and switch-case.

```

<!DOCTYPE html>
<html>
<head>
    <title>Simple Calculator</title>
    <script>
        function calculate() {

            // Get input values
            var num1 = parseFloat(document.getElementById("num1").value);
            var num2 = parseFloat(document.getElementById("num2").value);
            var operator = document.getElementById("operator").value;
            var result;

            // Check empty fields
            if (isNaN(num1) || isNaN(num2)) {
                alert("Please enter valid numbers");
                return;
            }

```

```

        // Switch-case for operations
        switch (operator) {
            case "+":
                result = add(num1, num2);
                break;

            case "-":
                result = subtract(num1, num2);
                break;

            case "*":
                result = multiply(num1, num2);
                break;

            case "/":
                if (num2 == 0) {
                    alert("Cannot divide by zero");
                    return;
                }
                result = divide(num1, num2);
                break;

            default:
                alert("Invalid Operator");
                return;
        }

        document.getElementById("result").innerHTML = "Result: " +
result;
    }

    // Functions for operations
    function add(a, b) {
        return a + b;
    }

    function subtract(a, b) {
        return a - b;
    }

    function multiply(a, b) {
        return a * b;
    }

    function divide(a, b) {
        return a / b;
    }
</script>
</head>

<body>

<h2>Simple Calculator</h2>

Enter First Number:
<input type="text" id="num1"><br><br>

Select Operator:
<select id="operator">
    <option value="+">+</option>
    <option value="-">-</option>

```

```
        <option value="*">*</option>
        <option value="/">/</option>
    </select><br><br>
```

Enter Second Number:

```
<input type="text" id="num2"><br><br>
```

```
<button onclick="calculate()">Calculate</button>
```

```
<h3 id="result"></h3>
```

```
</body>
```

```
</html>
```